As a prime example, in each of the five string theories there is an equation that is meant to determine the value of the theory's coupling constant. Currently, however, physicists have been able to find only an approximation to this equation, in each of the five string theories, by mathematically evaluating a small number of relevant string diagrams using a perturbative approach. Here is what the approximate equations say: In any of the five string theories, the string coupling constant takes on a value such that if it is multiplied by zero the result is zero. This is a terribly disappointing equation; since any number times zero yields zero, the equation can be solved with any value of the string coupling constant. Thus, in any of the five string theories, the approximate equation for its string coupling constant gives us no information about its value.

While we are at it, in each of the five string theories there is another equation that is supposed to determine the precise form of both the extended and the curled-up spacetime dimensions. The approximate version of this equation that we currently have is far more restrictive than the one dealing with the string coupling constant, but it still admits many solutions. For instance, four extended spacetime dimensions together with any curled-up, six-dimensional Calabi-Yau space provide a whole class of solutions, but even this does not exhaust the possibilities, which also allow for a different split between the number of extended and curled-up dimensions.6

What can we make of these results? There are three possibilities. First, starting with the most pessimistic possibility, although each string theory comes equipped with equations to determine the value of its coupling constant as well as the dimensionality and precise geometrical form of space time—something no other theory can claim—even the as-yet-unknown exact form of these equations may admit a vast spectrum of solutions, substantially weakening their predictive power. If true, this would be a setback, since the promise of string theory is that it will be able to explain these features of the cosmos, rather than require us to determine them from experimental observation and, more or less arbitrarily, insert them into the theory. We will return to this possibility in Chapter 15. Second, the unwanted flexibility in the approximate string equations may be an indication of a subtle flaw in our reasoning. We are attempting to use a perturbative